

Robustness of Vision Algorithms Under Image Degradations

Digital Image Processing — Course Project

Nadav Moshe · Alon Ron · 2025–2026

The Question

When dashcam images degrade, do **classical fixes** help — or do we need to **retrain the model**?

- **Distortions:** noise · low light · JPEG
- **Fixes tested:** enhancement (NLM / CLAHE / bilateral) vs fine-tuning
- **Tasks:** ORB · YOLOv8n · SegFormer-b0

Dataset — BDD100K

100 dashcam images with boxes + segmentation masks (seed 42).

BDD100K EDA — train split (3 samples)



Three Distortions, Four Levels Each

Distortion	Parameter	Levels (mild → severe)
Gaussian noise	SNR (dB)	30 · 20 · 10 · 5
Low light	γ	0.75 · 0.5 · 0.35 · 0.2
JPEG	Quality Q	90 · 50 · 20 · 10

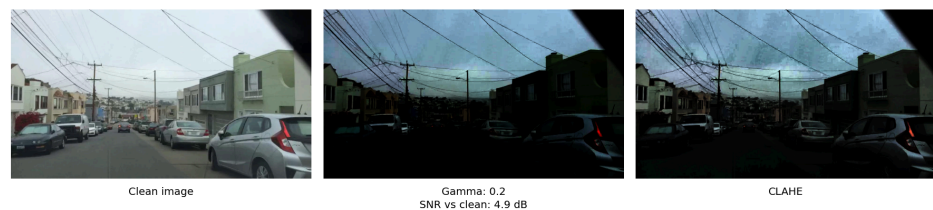
Distortion intensity sweep - 440ae5e7-00000000.jpg



Distorted vs Enhanced

Each distortion gets one matched enhancement (NLM · CLAHE · bilateral).

Distortion preview — 440ae5e7-00000000.jpg



Clean Baselines

Task	Metric	Clean score
ORB matching	Match ratio	1.000
YOLOv8n	Recall @ IoU 0.5	0.319
SegFormer-b0	Mean IoU	0.469

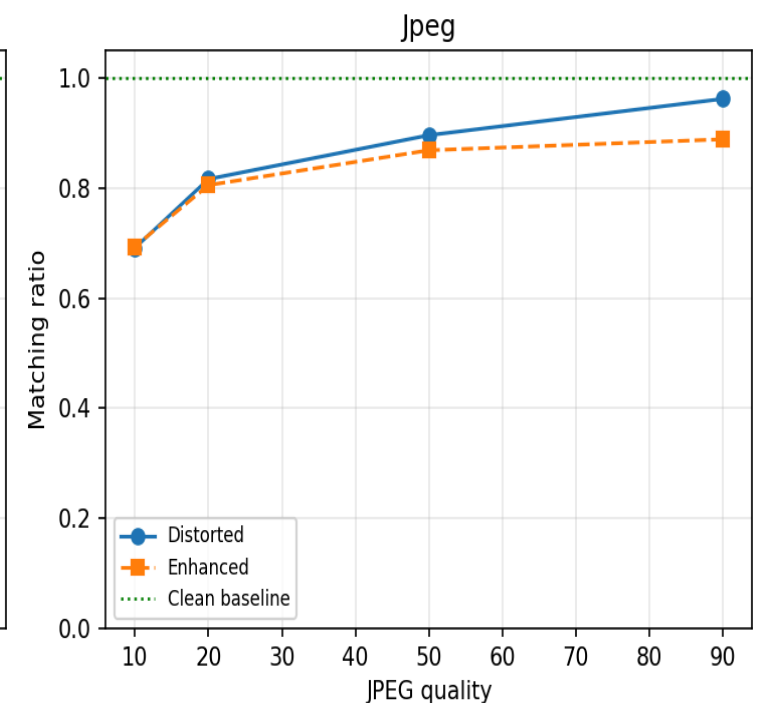
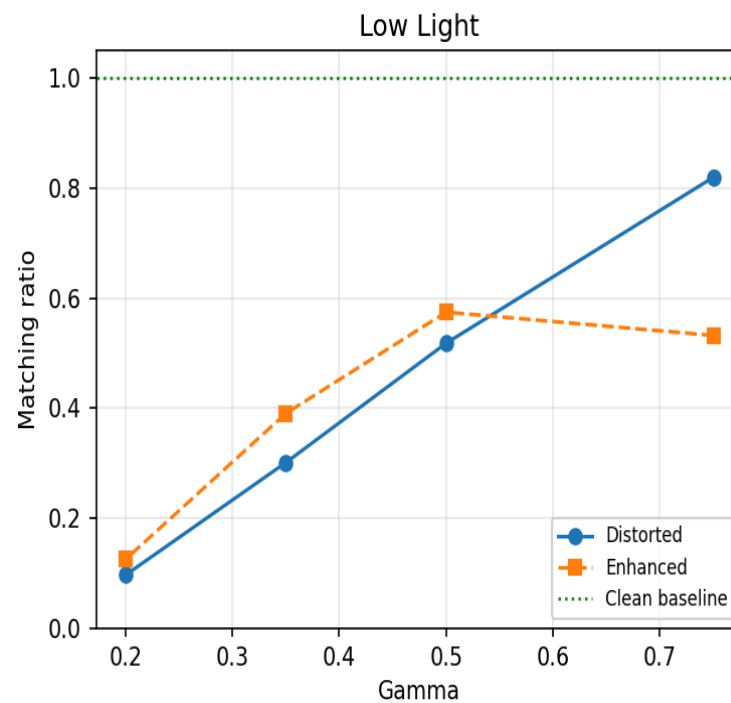
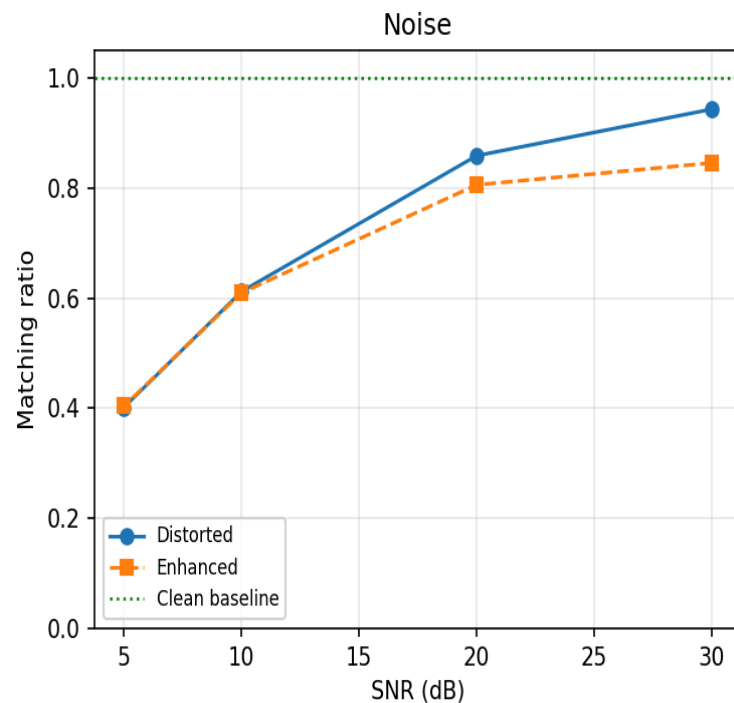
Trends on a 100-image subset — not official leaderboard scores.

ORB — Fragile in Low Light

Worst case $\gamma = 0.2$: matching ratio drops to **0.097**.

CLAHE helps low light (+0.089); NLM hurts mild noise.

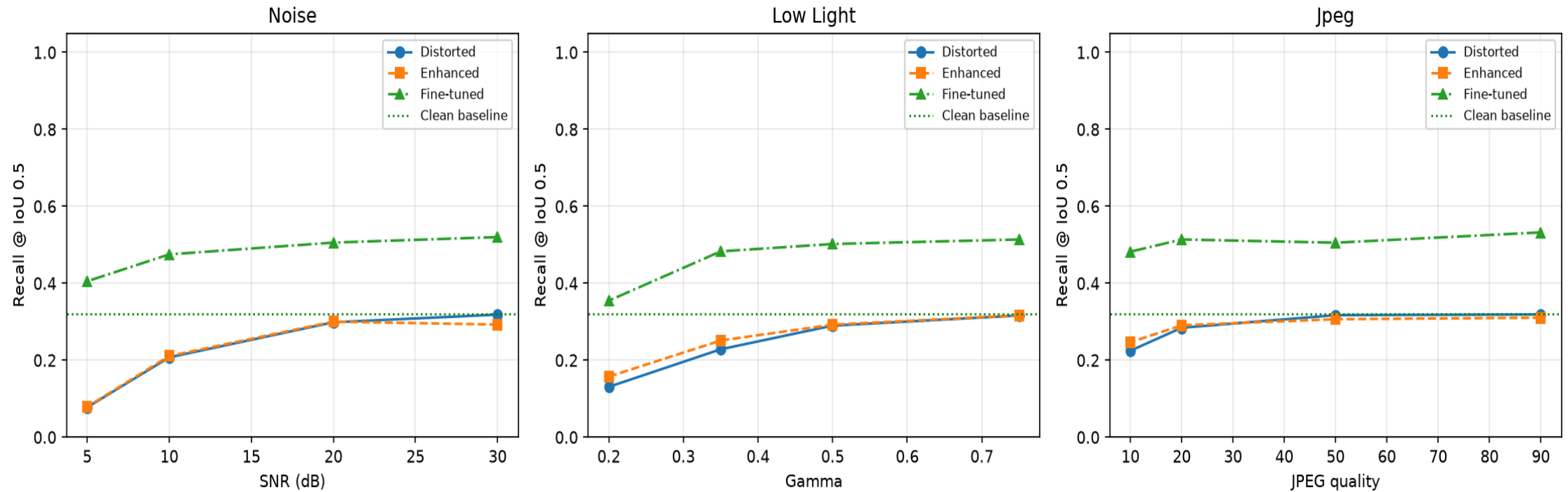
ORB matching ratio vs distortion intensity



YOLO — Collapses Under Noise

Recall at SNR 5 dB: **0.077** (from 0.319). Enhancement barely helps.

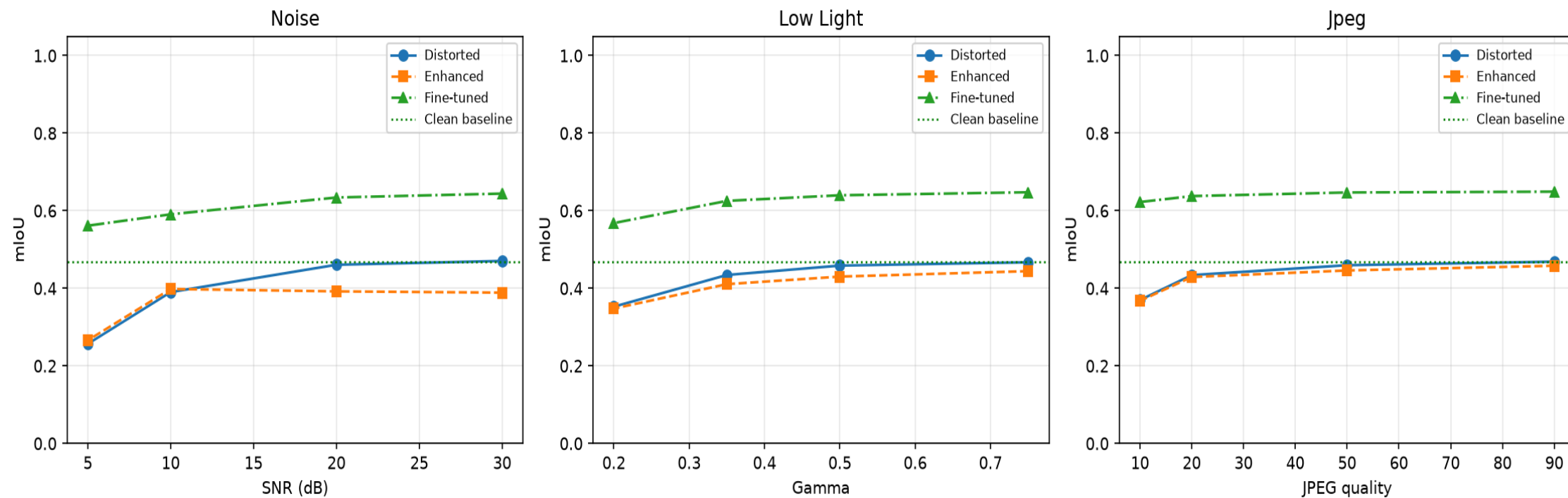
YOLOv8 detection recall vs distortion intensity



SegFormer — Noise Blurs Boundaries

mIoU at SNR 5 dB: **0.257** (from 0.469). NLM even hurts mild noise.

SegFormer mIoU vs distortion intensity



Key Insight

A cleaner-looking image is **not** a better input for the model.

- **NLM** smooths texture → hurts ORB and SegFormer
- **CLAHE** helps ORB contrast, not deep models
- **Bilateral** can't restore lost JPEG detail

Classical enhancement rarely recovers frozen deep models.

Fine-Tuning Fixes It

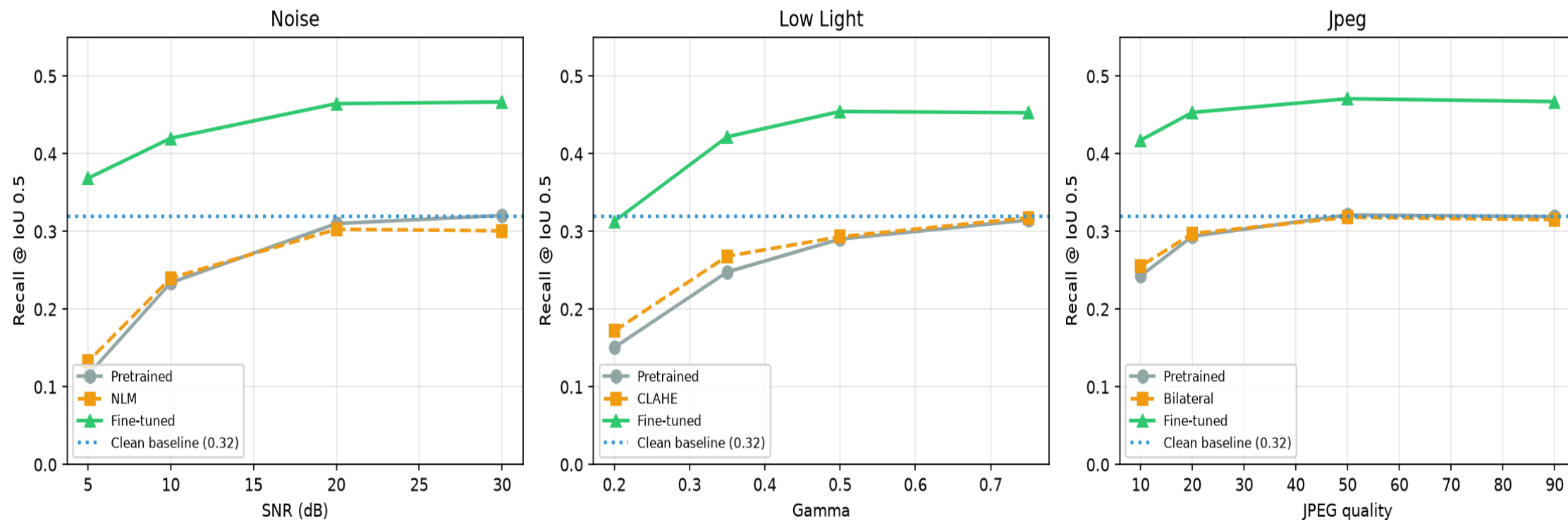
Retrain on distorted data (500 train / 100 val per level, 30 epochs).

Task	SNR 5 dB distorted	Fine-tuned
YOLO recall	0.077	0.405
SegFormer mIoU	0.257	0.561

Measured on the same 100 images as the frozen models.

Fine-Tuning — YOLO

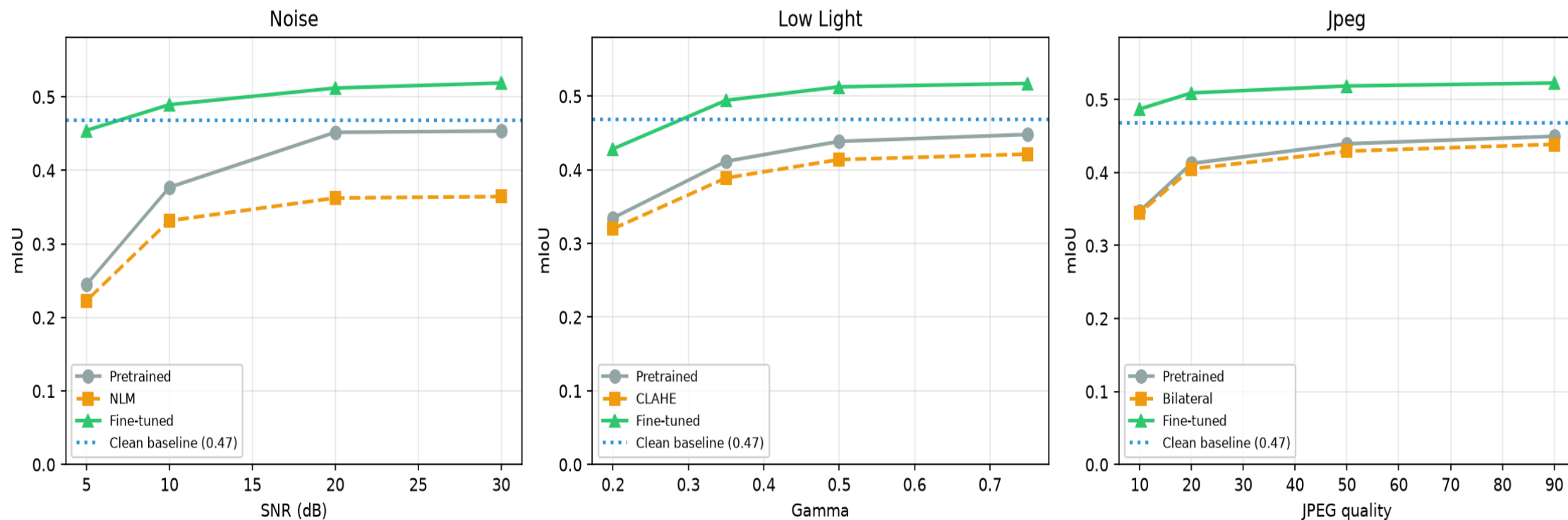
YOLO fine-tuning — recall vs distortion intensity
500 train / 100 val · 30 epochs



Fine-tuned recall beats frozen + enhanced at every level.

Fine-Tuning — SegFormer

SegFormer fine-tuning — mIoU vs distortion intensity
500 train / 100 val · 30 epochs



Fine-tuned mIoU beats frozen + enhanced at every level.

What Works When

Distortion	Best fix
Gaussian noise	Fine-tuning
Low light	CLAHE (ORB) · fine-tuning (deep)
JPEG	Fine-tuning

Conclusions

1. Frozen models fail under heavy noise; enhancement barely helps.
2. ORB sometimes gains from CLAHE; NLM often hurts.
3. **Fine-tuning is the reliable fix** for YOLO and SegFormer.
4. Low-level and high-level robustness diverge.

Thank You

github.com/NadavMoshe1/Image-Processing

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